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Data Engineering Career Roadmap

A practical learning plan for freshers and experienced professionals

Use this roadmap to build demonstrable data engineering skills in a sensible order. Freshers can follow the foundation path over about 20 weeks; working engineers can use the 12-week path to fill gaps and deepen one specialization.

The goal is a portfolio that shows you can build, validate, schedule and explain a data pipeline. A course or certificate cannot guarantee interviews or selection.

Choose your starting point:

If you are...	Start Here	Suggested weekly time
A fresher or career switcher	Build SQL and Python foundations first; then complete one end-to-end portfolio pipeline.	8–10 hours for about 20 weeks
Already working in data / software	Skip skills you can demonstrate. Choose one cloud platform and deepen reliability, scale, governance, or streaming.	6–8 hours for about 12 weeks

The core skill sequence :

Order	Skill	What you should be able to demonstrate
1	SQL and data modeling	Joins, CTEs, window functions, indexes, keys, normalization, star schemas.
2	Python, Linux, Git	Read APIs and files, write reusable code, use a terminal, commit work, handle errors.
3	ETL / ELT and warehouses	Ingest, transform, load, choose suitable storage and explain trade-offs.
4	Orchestration and data quality	Schedule dependencies, retry failures, log runs, add tests and data checks.
5	Cloud and distributed processing	Use one cloud and one warehouse; process larger data with Spark when needed.
6	Production practices	Document, monitor, secure, control cost, test changes and recover from failures.

Keep the toolset focused: learn one cloud deeply (AWS, Azure, or Google Cloud) and one warehouse/lakehouse first. Add a second stack only when a target role asks for it.

Fresher Roadmap :

Plan for roughly 20 weeks at 8–10 hours per week. Move faster if you already know SQL or Python. Every phase ends with something you can show in a GitHub repository.

Weeks	Learn	Build / prove it	Free learning
1–4	SQL foundations, Python basics, Git and terminal basics.	Query a public dataset; write a small Python script that reads CSV/JSON and handles missing values.	freeCodeCamp SQL and Python YouTube; SQLBolt practice.
5–8	Relational databases, schema design, ETL vs ELT, warehouse concepts.	Load an API or open dataset into PostgreSQL; design clean tables and document assumptions.	DataTalks.Club Zoomcamp data warehouse modules; PostgreSQL docs.
9–12	Python ingestion, APIs, Docker basics, orchestration with Airflow.	Create a repeatable pipeline with logging, retries and a scheduled DAG.	DataTalks.Club Zoomcamp; Apache Airflow tutorials.
13–16	Choose cloud (AWS / GCP / Azure), object storage, Spark or managed pipelines.	Run a batch pipeline on your chosen platform; explain storage and cost choices.	Databricks free training; cloud provider free learning paths.
17–20	Testing, data quality, dbt basics, monitoring, interview practice.	Polish an end-to-end project: source → raw → transform → tests → schedule → README and diagram.	dbt Learn Fundamentals; Zoomcamp final-project materials.

Fresher course options

Platform and course	Rating snapshot	Best fit	Fee guide
Coursera • IBM Data Engineering Professional Certificate Open course	4.6/5; 62K+ reviews (page snapshot)	Structured beginner path across SQL, Python, databases, ETL, Airflow/Kafka, Spark and projects.	Subscription; Coursera's public single-program price range is typically ₹4,650 - ₹7,490 /month. billing duration affects total.
Udemy • Data Engineering for Beginners: SQL, Python & Spark Open course	4.3/5; 8.9K ratings	Long, hands-on introduction to PostgreSQL, Python, PySpark and Databricks. Check the course preview and recent reviews before buying.	One-time purchase. Udemy sale and India checkout price vary; verify the amount shown before payment.
DataCamp • Associate Data Engineer in SQL Open track	Track rating not consistently shown on current listing	Short, interactive SQL + database modeling + ETL/ELT practice; useful as a guided skills track.	Included in DataCamp Premium. Public page lists ₹1,329/month billed annually (₹15,940 upfront equivalent); Basic is free but limited to first chapter of courses. Local checkout may differ.

Course ratings and review counts change. Treat them as learner feedback snapshots not a guarantee of course quality or hiring outcomes. Fees exclude any applicable taxes and cloud lab charges.

Experienced professional roadmap

Use a short skills audit first: list the systems you have shipped, the scale and service levels you owned, and the gaps that appear repeatedly in target job descriptions. Select one specialization rather than collecting tools.

Weeks	Focus	Work outcome	Free learning
1–2	Audit architecture and fundamentals: SQL performance, modeling, partitioning, incremental loads.	Draw one pipeline you own; identify bottlenecks, failure modes and measurable improvements.	DataTalks.Club course materials; database and cloud documentation.
3–5	Scale and lakehouse: Spark execution, file formats, partitioning, skew, batch vs streaming.	Benchmark a workload; document before/after runtime, cost, or data volume.	Databricks free Data Engineer training and Spark documentation.
6–8	Reliability: Airflow orchestration, idempotency, retries, backfills, tests, observability.	Build a failure drill: simulate a bad input, alert, recover safely, and capture run history.	Apache Airflow tutorials; dbt Learn; OpenLineage learning materials.
9–10	One specialization: cloud data engineering, streaming, governance, or platform engineering.	Create a small reference implementation and an architecture decision record.	Provider learning path (AWS / Google Cloud / Azure); Kafka docs if streaming.
11–12	Portfolio and interview readiness: trade-offs, incidents, security, cost and mentoring.	Publish a sanitized case study with architecture, SLAs, risks and measurable outcomes.	Revisit course labs selectively; practice system-design explanations.

Experienced course options

Platform and course	Rating snapshot	Best fit	Fee guide
Coursera • Preparing for Google Cloud Certification: Cloud Data Engineer Open certificate	4.6/5; 4.9K reviews	GCP-focused refresh: BigQuery, data lakes/warehouses, batch and streaming pipelines; intermediate.	Subscription; public single-program range typically USD $49–$79/month. India price shown at checkout.
Udemy • Complete Hands-On Introduction to Apache Airflow 3 Open course	4.5/5; 14.6K ratings	Focused orchestration upgrade: DAGs, tasks, executors, Docker and production workflow concepts.	One-time purchase. Public listing showed a sale price of USD $11.99 and crossed-out USD $79.99 when checked; offer and India price vary at checkout.
DataCamp • Data Engineer in Python Open track	4.3/5; 52 reviews	40-hour guided track covering Python, ETL/ELT, Airflow, Git, tests and pipeline work. Requires SQL foundations.	Included in DataCamp Premium. Public page lists USD $14/month billed annually; annual fee is charged upfront. Local price may differ.
Udemy • Data Engineering Project: SQL, Python, Airflow, Docker, CI/CD Open course	4.4/5; 750+ ratings	Project-based practice for turning APIs and SQL into a tested, containerized pipeline.	One-time purchase; dynamic sale and India checkout price. Check recent reviews and update date before enrolling.

Experienced learners can use these courses as targeted refreshers. Confirm that the tools and cloud services still match the roles you are pursuing.

Free YouTube and learning platforms :

Start with free material and build projects before paying for a certificate. Use the playlists and official learning pages below as the backbone of either roadmap.

Resource	Cost	Use it for
DataTalks.Club Data Engineering Zoomcamp 2026 YouTube playlist • Course page	Free; self-paced materials available. Live cohort announced for Jan 2027.	End-to-end project learning across ingestion, warehouses, orchestration, Spark, streaming and a final project. Basic SQL and coding help.
YouTube channel : freeCodeCamp SQL and Python courses	Free on YouTube	Use for beginner refreshers in SQL, Python, command line and Git. Choose a recent course and practice alongside it.
Youtube channel **Data Engineering Fundamentals for Analysts (Codebasics)**	Free on Youtube	Covers the data engineering fundamentals every data analyst needs to know to start building real pipelines , not just dashboards.
Databricks training Free training hub	Free learning options; some labs may need an account	Practice lakehouse and data engineering fundamentals, including SQL, Python and pipeline concepts.
dbt Learn Fundamentals Free courses	Free self-paced learning	Learn models, sources, tests, documentation and deployment basics for analytics engineering.
Apache Airflow tutorials Official docs	Free	Learn DAGs, tasks, scheduling and core orchestration patterns from project documentation.
SQLBolt Interactive lessons	Free	Practice SQL concepts in short browser-based lessons; pair it with a database project.

One portfolio project that works for either path

- Choose a public dataset or API with a clear question such as daily transit, weather, public finance or retail data.
- Ingest raw data with Python and store an untouched copy. Add schema checks and useful logs.
- Load to PostgreSQL or a cloud warehouse; model clean tables and transform with SQL or dbt.
- Schedule the pipeline with Airflow or a managed service. Make reruns safe and document recovery steps.
- Add tests for nulls, duplicates, freshness and row counts. Capture at least one failure and its fix.
- Publish a README, architecture diagram, sample output, assumptions, cost notes and a short demo.

Course and fee notes

- Prices and discounts are dynamic. The fee column reflects public platform pages checked on 15 October 2026; India checkout, taxes, coupons, subscriptions and app-store pricing can differ.
- Coursera programs may be billed as a monthly subscription. Finish efficiently and check the renewal date; the total depends on how many months you remain enrolled.
- DataCamp's public Premium page showed USD $14/month billed annually, with the full annual charge upfront. Basic access includes only the first chapter of each course.
- Udemy individual course prices vary by region and promotion. Verify the final India price and course update date at checkout.
- Free learning can still incur cloud charges if you create paid resources. Use free tiers carefully, set budgets/alerts, and delete test resources when done.

Reference links

- DataTalks.Club Zoomcamp course documentation
- Databricks training
- dbt Learn
- Coursera pricing and program information
- DataCamp pricing
- DataCamp Associate Data Engineer in SQL
- DataCamp Data Engineer in Python
- Udemy Data Engineering for Beginners: SQL, Python & Spark
- Udemy Airflow 3 hands-on course
- Udemy project-based data engineering course
- Coursera IBM Data Engineering certificate
- Coursera Google Cloud Data Engineer certificate
- Apache Airflow tutorials

Prepared as a learning guide, not a hiring guarantee. Role requirements vary by company, region and seniority; compare this plan with current job descriptions and prioritize repeated requirements.